Trees 3 (Euler Tour)

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Expert at codeforces (1817)
5 star at codechef (2040)

Euler Tour (Tree Flattening)

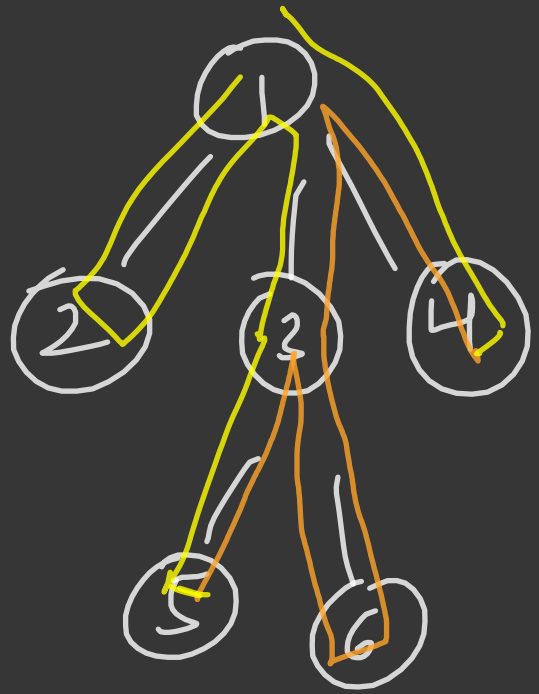
- Euler Tour (2 Types)
- Questions
  - Subtree Queries
  - Path Queries
  - Tree Queries

Euler Tour (segment Tree)

→ 30 mins (LCA question)

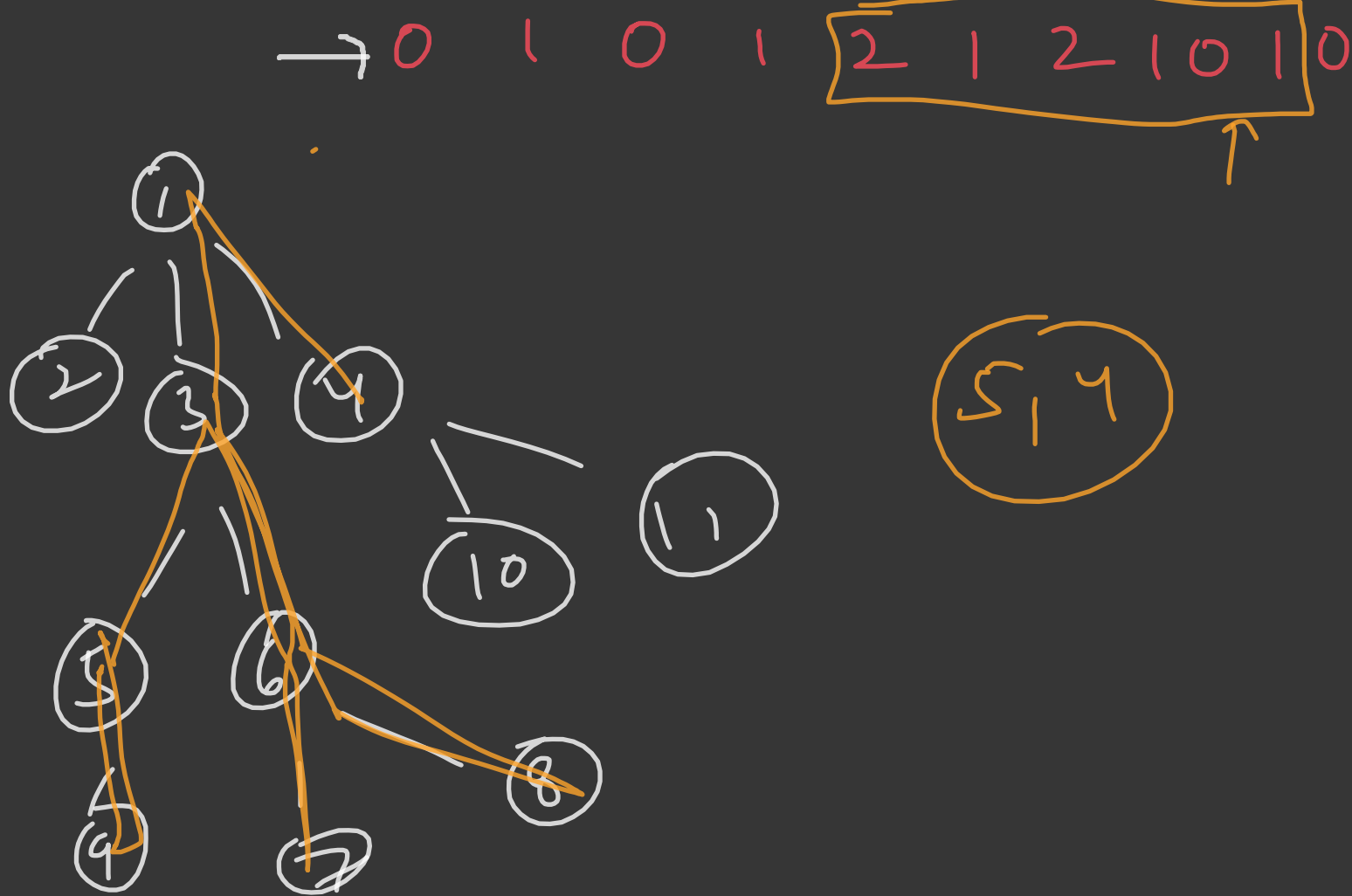
Euler Tour →

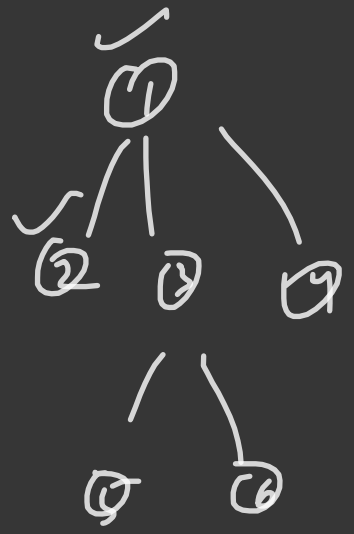
Type 1



lca (5, 4) = ?

Flat → 1 2 1 3 5 3 6 3 1 4 1


0 1 0 1 2 1 2 1 0 1 0
1
2
3
4
10
11
5
6
8
9
7
5 6

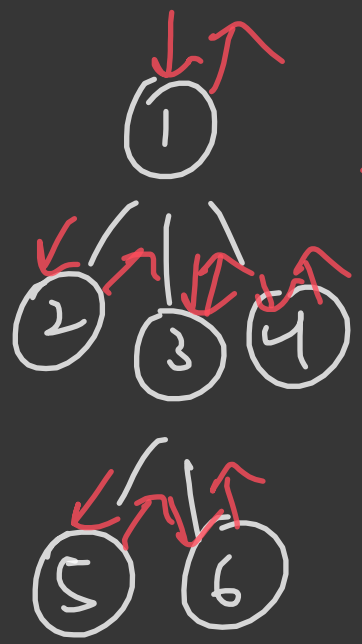


2 2 2

$TC \rightarrow O(N) + O(9 \times \log N)$

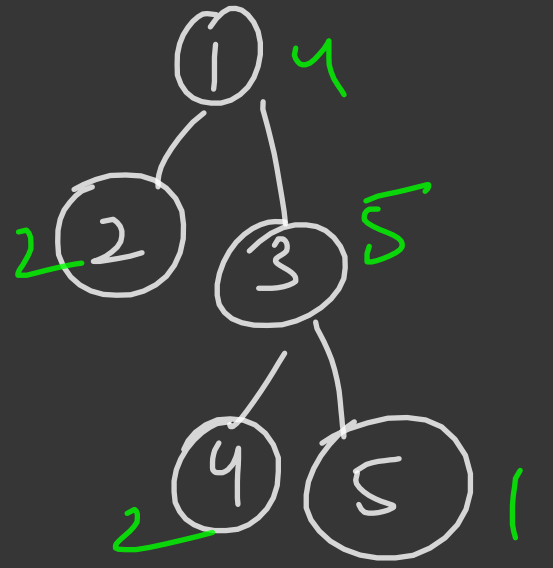
$N \log N$ — AC

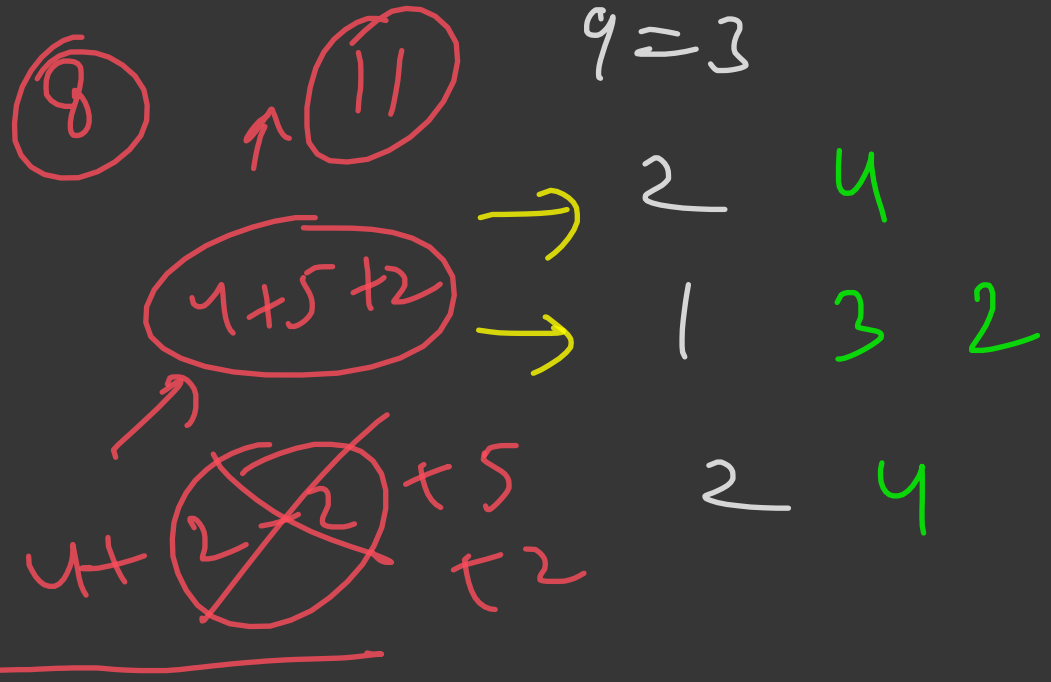
Type2



flat → 1 2 2 3 5 5 6 6 3 4 4 1

Q

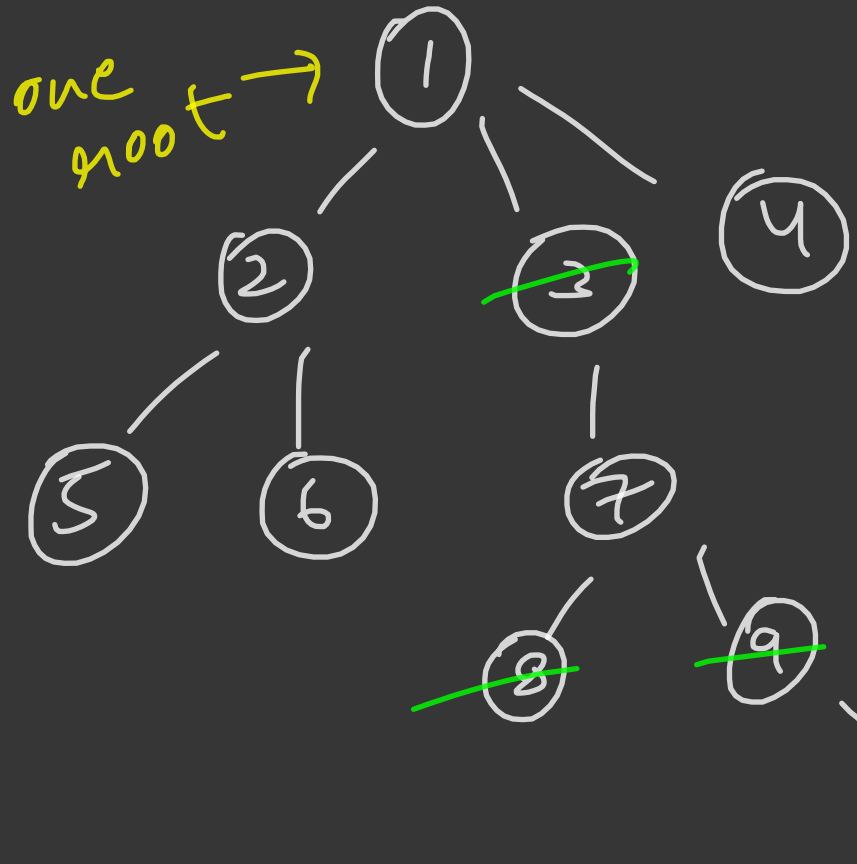




flat → 1 2 2 3 4 4 5 5 3 1

segan → 4 2 -2 2 2 -2 3 -3 -2 -4

Q



[9 → 3, 8, 9, 10]

lowest level node → ?

node